

PRACTICAL 8: VLAN IMPLEMENTATION AND NETWORK SEGMENTATION

Objective

Create VLANs and segment network traffic.

Equipment

- Cisco Catalyst 3560 Switches
- 10 PCs

VLAN Design

VLAN	Department	Devices
VLAN 10	Administration	PC1-PC3
VLAN 20	Lecturers	PC4-PC6
VLAN 30	Students	PC7-PC10

Activities

- 1.Create VLANs.
- 2.Assign ports to VLANs.
- 3.Verify VLAN membership.
- 4.Test communication.
- 5.Document VLAN design.

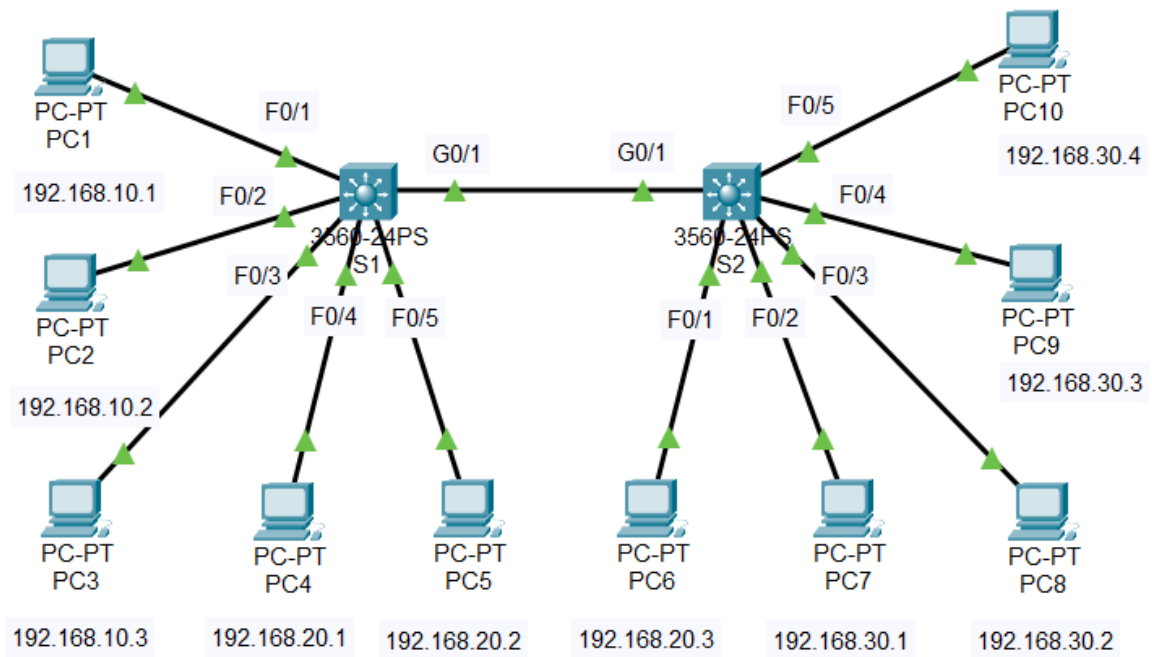
Example Commands

- vlan 10
- name ADMIN
- vlan 20
- Name LECTURERS
- vlan 30
- name STUDENTS

Verification

- show vlan brief

Topology:



Activities

Create VLANs.

Switch 1

 S1

Physical

Config

CLI

Attributes

IOS Command

```
Switch(config)#  
Switch(config)#vlan 10  
Switch(config-vlan)#name Administration  
Switch(config-vlan)#exit  
Switch(config)#vlan 20  
Switch(config-vlan)#name Lecturers  
Switch(config-vlan)#exit  
Switch(config)#vlan 30  
Switch(config-vlan)#name Students
```

Switch 2

 S2

Physical

Config

CLI

Attributes

IOS Command

```
Switch>enable  
Switch#confi ter  
Enter configuration commands, one per line. 1  
Switch(config)#vlan 10  
Switch(config-vlan)#name Administration  
Switch(config-vlan)#exit  
Switch(config)#vlan 20  
Switch(config-vlan)#name Lecturers  
Switch(config-vlan)#exit  
Switch(config)#vlan 30  
Switch(config-vlan)#name Students
```

Assign ports to VLANs.

VLAN 10

```
S1
Switch(config)#interface f0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface f0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface f0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#
```

VLAN 20

```
S1
Switch(config)#
Switch(config)#interface f0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface f0/5
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#
```

```
S2
Switch(config)#interface f0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#
```

Vlan 30

```
S2
Switch(config)#interface f0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#interface f0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#interface f0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#interface f0/5
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access van 30
Switch(config-if)#
% Invalid input detected at '^' marker.

Switch(config-if)#switchport access vlan 30
Switch(config-if)#
```

Trunk between switches

 S1

```
Switch(config)#interface g0/1
Switch(config-if)#switchport mode trunk
Command rejected: An interface whose trunk encapsulation is "Auto" can not be
configured to "trunk" mode.
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#
```

 S2

```
Switch#confi ter
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface g0/1
Switch(config-if)#switchport mode trunk
Command rejected: An interface whose trunk encapsulation is "Auto" can not be
configured to "trunk" mode.
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#
```

Verify VLAN membership.

Switch 1

S1

```
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24, Gig0/1 Gig0/2
10	Administration	active	Fa0/1, Fa0/2, Fa0/3
20	Lecturers	active	Fa0/4, Fa0/5
30	Students	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

Switch#

Switch 2

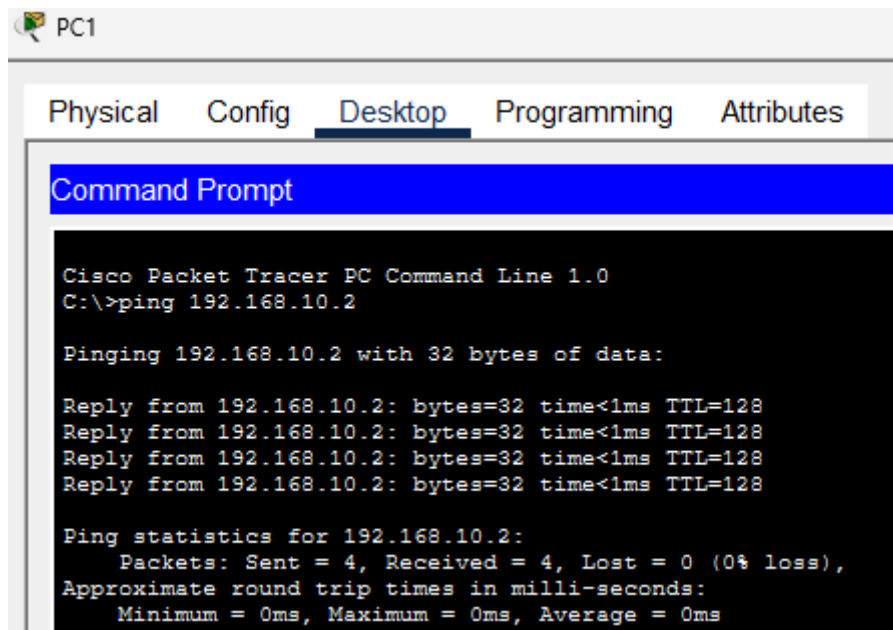
S2

```
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24, Gig0/1 Gig0/2
10	Administration	active	
20	Lecturers	active	Fa0/1
30	Students	active	Fa0/2, Fa0/3, Fa0/4, Fa0/5
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

Test communication.

Ping the same vlan e.g. vlan 10



PC1

Physical Config **Desktop** Programming Attributes

Command Prompt

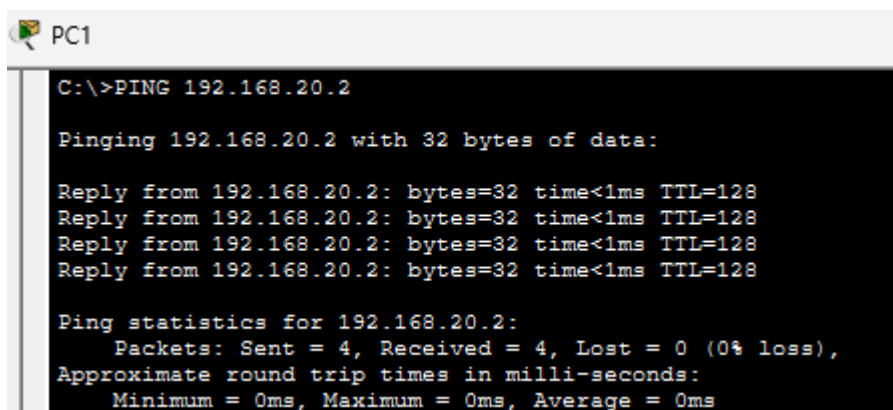
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Ping from vlan 10 to vlan 20 (PC3 to PC5)



PC1

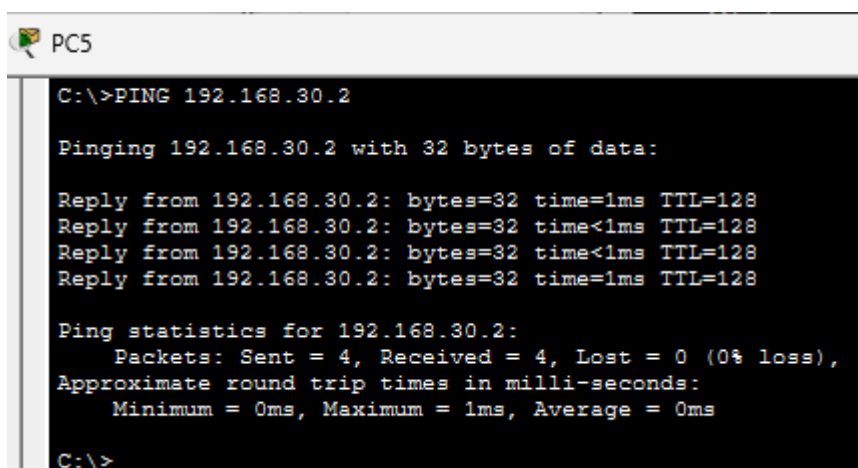
```
C:\>PING 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Reply from 192.168.20.2: bytes=32 time<1ms TTL=128
Reply from 192.168.20.2: bytes=32 time<1ms TTL=128
Reply from 192.168.20.2: bytes=32 time<1ms TTL=128
Reply from 192.168.20.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

PING VLAN 20 TO 30



PC5

```
C:\>PING 192.168.30.2

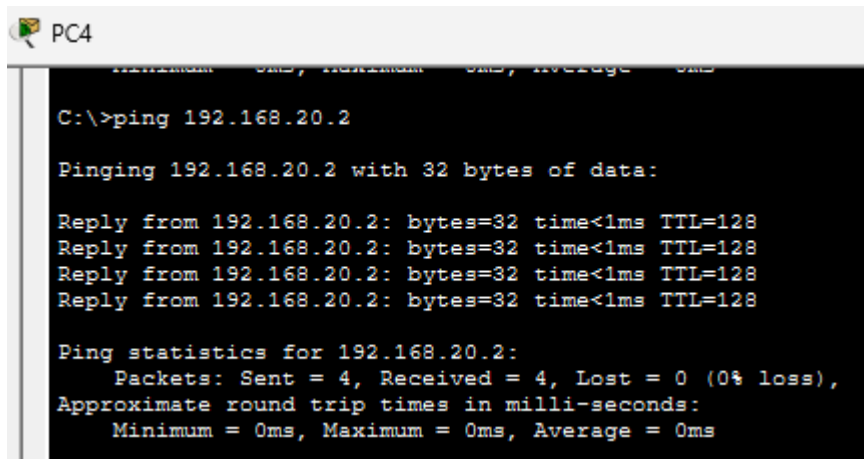
Pinging 192.168.30.2 with 32 bytes of data:

Reply from 192.168.30.2: bytes=32 time=1ms TTL=128
Reply from 192.168.30.2: bytes=32 time<1ms TTL=128
Reply from 192.168.30.2: bytes=32 time<1ms TTL=128
Reply from 192.168.30.2: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.30.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Ping inside vlan 20:



```
PC4
Minimum    0ms, Maximum    0ms, Average    0ms

C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Reply from 192.168.20.2: bytes=32 time<1ms TTL=128
Reply from 192.168.20.2: bytes=32 time<1ms TTL=128
Reply from 192.168.20.2: bytes=32 time<1ms TTL=128
Reply from 192.168.20.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Document VLAN design.

VLAN 10: VLAN 10 was assigned to PC 1, 2 and 3 on switch 1

VLAN 20: VLAN 20 was assigned to PC4 and 5 on switch 1 and PC6 on switch 2

VLAN 30: VLAN 30 was assigned to PC 7,8,9 and 10 on switch 2

Deliverable

Working segmented network.

Assessment Rubric (15 Marks)

Criteria	Marks
VLAN creation	4
Port assignment	4
4 Connectivity verification	4
Documentation	3
3 Total	15